A new handling system for textile/clothing industry

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Abstract

The paper proposes a robotized handling system able to grasp and handle limp and soft near 2D materials like natural and synthetic fabrics. The robotic manipulation of these difficult objects without a own defined shape is a needed step to extend automation beyond the fabric parts cutting in clothing and technical garment manufacturing environments by exploiting options of robotics, already in use for most of the manufacturing of automotive electric, electronic and mechanical goods. The handling system was conceived with the objective of grasping parts from the cutting table and forward them singled out and kept flat in a well defined pose to the following manufacturing sections. The robotic system purposely developed is presented: it includes: - a reconfigurable passive hanger; - an innovative vacuum picking module; - a metamorphic robotic gripper. The system architecture and main components are presented. The grasping system was realized and tested within a pilot plant in industrial premises.

1 Introduction

The textile industries in the last years are crossing, especially in Europe, a period of a deep crisis. For this reason the European Community focused its attention on this sector to induce the research community to develop new technologies able to help the textile and clothes industry. The low level of automation of the textile industry is caused principally by the particular characteristics of the material that is handled during the work process. The fabric, in fact, as all limp materials, is very difficult to grasp and handle, and at the same time requires attention and forethought in the cut, transport, sewing and finishing operation [1]. So, high human-labour intensive sectors such as clothing and footwear manufacturing are addressing flexible automation and robotization but the lack in available technology and systems for adaptive handling of limp, synthetic and natural materials is one of the main hindrances. Several handling operations occur along the manufacturing process of these goods. The majority of these operations today is carried out by manual labour and none of the past at-tempts at innovating the overall manufacturing system by introducing new handling technologies were successful and applied in industry.

Typical handling tasks are picking and orderly placing of fabric parts. Main concerns are: robust and reliable grasping of limp parts without damages and accurate positioning for the following manufacturing operations guaranteeing the quality of the final product. The development of robotic systems to carry out tasks involving soft and limp material is challenging and few reliable systems exist most of which narrowly oriented to specific tasks [2], [3],[4]. Skills involved may comprise tactile and vision although material properties can be taken into account to achieve intrinsic grasping robustness [5].

This paper presents the research and implementation work carried out in the frame of the Leapfrog EU Integrated Project [6].

Cutting table unloading is considered a reference task because it is a necessary step in an automated garment manufacturing process today and it involves all main handling issues: picking from flat a thin soft part without own stable shape, lifting up, displace, deliver to further manufacturing operations with required position accuracy and complying with quality requirements. The development case considered is the production of jackets for formal men's suits. A fully automatic cutting table unloading cell has been developed and was industrially demonstrated. As part of it, the new concept handling system composed by a metamorphic handling device and a reconfigurable hanger has been realised. The metamorphic handling device is a robotic branched unit able to reconfigure in order to adapt to the fabric part different shapes and material [7]. The hanger is a passive device whose three fingers are actuated by the handling device and three fingertips clamps are fed by the picking modules embedded into the handling device. The hanger task is to hold the individual fabric item during transportation to the following manufacturing stations.

The locations of the areas of interaction between the handling system and material assigned to special picking modules applied on the handling device fingertips and driven by its fingers. Active adhesion to the material at the areas of interaction can be realized by dedicated devices using alternative physical principles e.g. vacuum, electrostatic, freezing.

The paper presents a solution encompassing several new concepts for fabric handling in the sector of garment manufacturing. A robotic reconfigurable gripper and a method for the application of reconfigurable hangers to the fabric parts manipulated is presented. The robotic handling

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system has been realized and tested within the pilot plant in Castelletto Ticino (Italy) in summer 2009.

2 New handling system concept

The robotic system is developed to carry out automatic unloading of fabric parts laid down in single layer on the stiff flat surface of a cutting table. The cut parts can be made of different material: wool, linen, cotton and artificial fibres, can have various shape, are very thin, soft and are generally porous. Attention has to be given to possible damages to the high quality garment costly material during picking and handling tasks.
The cut parts, once singularly picked, need to be forwarded singularly in order to the following manufacturing sections so that further manual or automatic part manipulations are reduced at minimum. The concept applied by the authors is to circulate the parts within the manufacturing plant suitable hung on a given number of points and kept flat with the help of gravity.
From a large investigation on garment industry, taking into account the part shapes, three picking points positioned along one edge of the part appeared a good compromise to prevent folds with a not too complex hanger architecture. **Figure 1** shows the picking point positions defined for some man jacket parts.

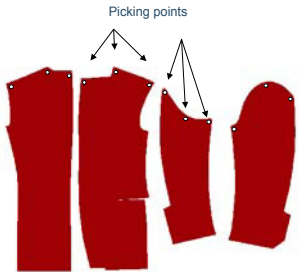


Figure 1 Example of picking point reference triads for some jacket panels.

The positions of the picking points on each part are given from the CAD nesting station: their computation rely on 3D CAD/CAE software specific for near 2D soft parts stress distribution analysis.
Taking into account the before mentioned requirements, the robotic system has to present the following basic features for the unloading of the cutting table and circulating the cut parts to the operative manufacturing resources without loosing the achieved part shape/configuration.
Because each part needs a hanger with 3 clamps that circulate it within the plant, a great number of hanger is required and it was decided to realize low cost passive hangers that are not able to reconfigure by themselves but that can be reconfigured and operated by a more sophisticated robotic device.
A new grasping device able to perform these different task functions is developed as a multifunctional metamorphic robotic device [8] with good dexterity and suitable static and dynamic performances: it is endowed with continuous mobilities in order to be able to self adjust to the prescribed picking points triads and with some discrete on/off degree of freedom driving the hanger docking and parts picking and release functions. The grasping device is carried out by an industrial full mobility robotic arm able to move it and control its orientation. Robot, grasping device and hanger cooperate to perform the task.
A critical operation is the lifting of the porous soft material from the stiff surface of the cutting table on wich it lays down. Different physical principles were investigated and comparatively evaluated for this specific application: a new vacuum based picking module was conceived and developed.
A modular design approach was adopted in order to better exploit the innovative devices promoting their use in a wider set of industrial application.

3 Handling system design

The main components of the handling system are: - the *hanger*, able to circulate the part; - the *picking module* able to lift up the parts from the cutting table; - the *grasping device*, able to reconfigure itself and the hanger and to enter the part edge in the hanger clamps. This section presents the main design issues for these components. The reconfigurable gripper is carried and operated by a commercial industrial robotic arm.

3.1 Reconfigurable hanger

The adopted total life cycle design approach in a mass customized production suggested to use only one typology of hangers able to adapt to large sets of parts.
The parts of the target garment were analyzed with the objective of realizing a robotic system to fully automate their handling and the application of the hanger: based on this analysis we decided to use the same hanger for all the parts. This device is expected to be passive (apart the clamps which should release the part under some actuation signal) and to have a simple low cost architecture. Moreover, it should be easy to operate with enough robustness to avoid fall of the part and generation of errors (inaccuracy or slipping in the reconfigurable system carrying the clamps).
The hanger, **Figure.2**, comprises a body, three fingers each ending with a clamp and a mechanical-signal flange for interfacing to external devices and transportation system. A long handle on the body is available for manual handling especially when carrying a cut part along a manufacturing environment that foresees human assisted operations.

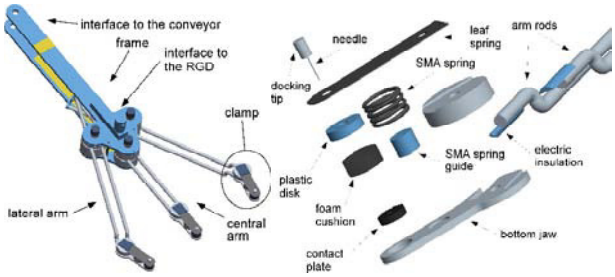


Figure 2 Digital mock-up and components of the hanger

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Each lateral finger has two passive freedoms: one rotation and one translation in direction orthogonal to the axis of rotation. A friction joint with grooves is adopted, with a preloading spring and a button to compress the spring and unlock both fingers freedoms at one time when pushed. The central finger has one translational freedom in direction orthogonal to the revolute joint axes of the lateral fingers, which are parallel each other. As a result the clamps of the three fingers move in the same plane.

3.2 Picking module

The picking module is designed as a holonic module providing the functionalities of picking and lifting up of the fabric. Grasping is provided by a subunit composed of an aluminium pipe with, at one extremity, the grasping mouth and at the other side a fan air flow generator complete of electrical motor. The aluminium pipe slides through the frame of the module and is supported by one side of a double fork link. At the other side, this link carries a hollow conic pipe whose bottom extremity envelopes the docking tip of the hanger clamp. The frame of a micro pneumatic cylinder is connected at the centre of the fork; The rod of this cylinder is screwed to the frame of the picking module. By this assembly the fork can be shifted vertically respect to the frame of the device: this results in the parallel shift of the vacuum pipe and of the hanger clamp docking pipe. A compliant elastomeric ring and a net cap are adopted to preserve fabric from damaging and to prevent it to be sucked inside the pipe.
With the picking module docked to one hanger clamp the grasping device finger can drag the corresponding hanger finger (with friction joint released) to the part grasping location. In this way the grasping device can fully reconfigure a hanger as a part of the unloading working cycle.

3.3 Reconfigurable grasping device

The handling strategy prospected asks for a multifunctional grasping device [9]. Required functionalities are: - docking to the frame of the hanger, - unlocking the passive joints of the hanger and resetting it, - operating the passive joints of the hanger to set the locations of the picking points, -lifting the cut part from flat by the picking modules, - commanding the opening of the clamps of the hanger, - inserting the lifted part into the clamps at the desired picking points locations and commanding their closure, - releasing the body of the hanger once moved back to the conveyor. The large handling maneuvers like reaching the free hanger and orienting the grasping device so that it can dock to it and the reverse procedure will be executed by the robot carrying the reconfigurable grasping device.
The reconfigurable grasping device, **Figure 3** complying with these remarks comprises three articulated fingers, two lateral 2 dof and one central 1 dof, with serial architecture and actuators distributed along the chain. Each finger carries at its end a grasping-picking module. The central finger comprises one prismatic joint and the corresponding picking module translates in a fixed direction. Each lateral finger contains three revolute joints, two active and one passive at fingertip, with parallel axes. A suitable mechanical architecture constrains the picking modules to translate in a plane orthogonal to the revolute joints of the fingers.

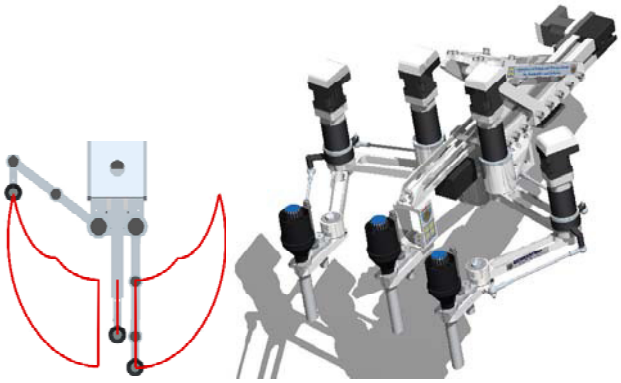


Figure 3 Grasping device workspace and digital mock-up

Kinematic, dynamic analysis and computational simulations were performed in order to assess the grasping device parameters. Typical results are shown in **Figure 4**.

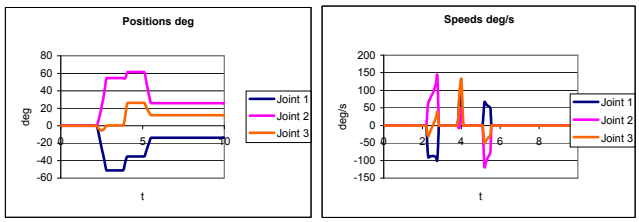


Figure 4 Dynamic simulation results for lateral finger joints position [deg] and velocity [deg/s] versus time [s]

The docking of the grasping device to the hanger body is made by a groove and a pneumatic lock guarantees for accurate self positioning and blockage of the hanger. Three dedicated pneumatic cylinders are used to release the friction joints of the hanger fingers.

3.4 Working cycle and system control

The table in Figure 5 illustrates the main steps of the handling system work cycle.
As first, the gripper docks to one hanger at a know location. When this operation has been completed, the grasping device blocks to the hanger through a locking mechanism including a cylinder and a conic peg.
The first operation done is the hanger reset. The robot rotates the grasping device with the central gripper finger vertical downwards. The three unlocking cylinders in the gripper are activated and they press and release the joints of the hanger; the fingers of the hanger fall down on their stops. Then the unlocking cylinders of the grasping device are deactivated and the fingers locked.
Now the secondary docking step starts. The cylinders on the picking module are activated; then the gripper fingers move the picking modules to the prescribed locations. The hollow pipes of the picking modules envelope the docking tips.

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The third step is the setting of the hanger. The hanger joints are released. The gripper fingers move the clamps to the desired grasping points on the part. The gripper deactivates again the releasing cylinders and the hanger joints block.
The fourth step is the grasping of the part. The service robot moves the grasping device on the part with the mouths of the vacuum pipes close to the desired grasping points. The cylinders of the picking module are still deactivated. When the mouths touch the part, vacuum is activated and the grasping forces generated.
The cylinders in the picking module are activated and they move up together with the grasped part. In the meantime the SMA springs in the clamps are activated and clamps opened. The hollow pipes of the picking modules move up. As soon as the clamps are open, the gripper fingers move the picking module backward inserting the fabric into the clamps. Then clamps are closed and the part is grasped.

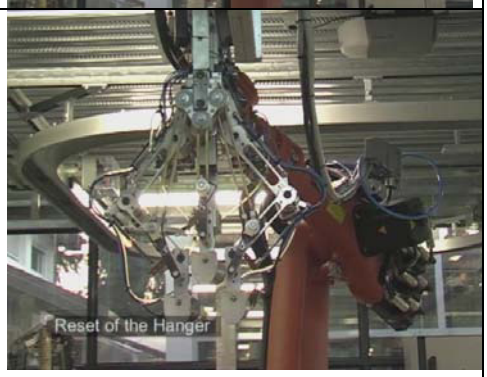
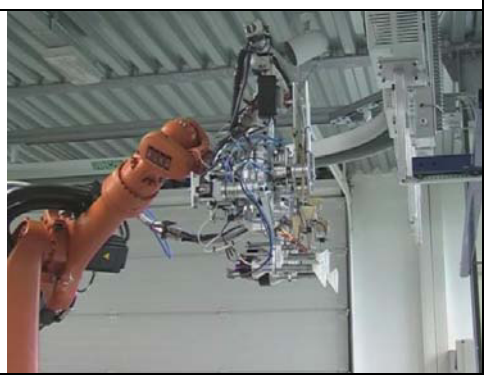
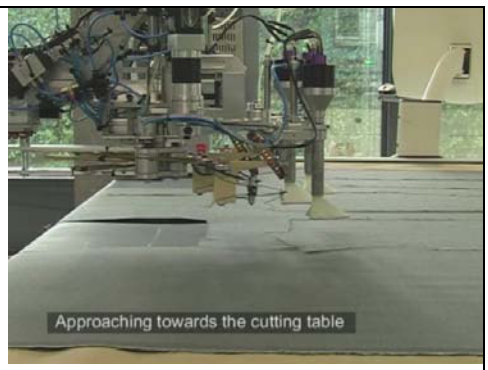
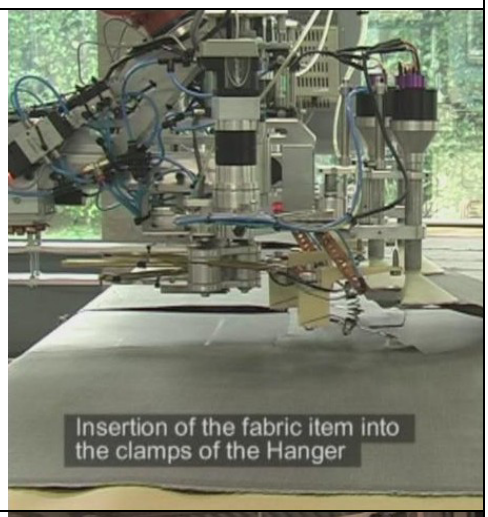
Step 1 GD docks to the H's frame at a known location	
Step 2 GD resets the H unlocking its passive joints. The H fingers stop in reset nominal position. GD matches the H fingertips with its GMs	
Step 3 GD and H are coupled and GD drives the reconfiguration process to the set-up required for grasping next part	
Step 4 The robot moves GD+H on the part so that the PMs are on the picking points. Vacuum is actuated and picking forces applied to the part	
Step 5 H clamps are opened, the GD moves backward the fingertips grasping the part and insert it between the clamps. The clams close	
Step 6 The assembly GD+H+ part is moved by the robot to the delivery location. The GD unlocks the H+part and releases it	

Figure 5 Steps in handling cycle. H= Hanger, GD= Grasping Device, PM= Picking Module.

4 Handling system prototyping and testing

Hereafter some brief notes on the main components realization are given.

4.1 Reconfigurable hanger prototype

The hanger palm and fingers are realized in aluminium. Each clamp is long and thin to realize a double finger-tip grasp; it comprises a base with a rubber pad at the tip and a leaf spring with a needle pressing the clamped part against

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the rubber pad and locking it in position by means of the needle. A shape memory alloy (SMA) spring actuator is adopted to lift up the leaf spring and open the clamp [10]. A very soft foam cushion around the needle helps the fabric to slide out and be released by the clamp when open. Electrical current flows through the SMA spring along the two sticks of which the hanger finger is made. Circuitry embedded in the hanger body connects all clamp actuators to the signal interface in the handle. Electrical current is provided to the hanger through this interface to open each single clamp.

4.2 Picking module prototype

The physical prototype of the picking module was realized in accordance with the virtual prototype including the pneumatic actuator commanding bottom plate translation. It is shown in **Figure 6**.

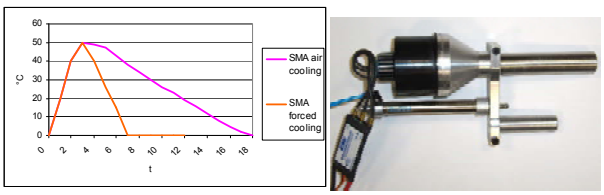


Figure 6 Picking module prototype

4.3 Robotic grasping device prototype

The robotic grasping device is designed as 3 fingers symmetric hand with 2 lateral identical fingers and a different central one.

4.3.1 Lateral fingers

Each lateral finger is composed of one palm frame (to be connected to the central linear actuator), and 3 phalanges.

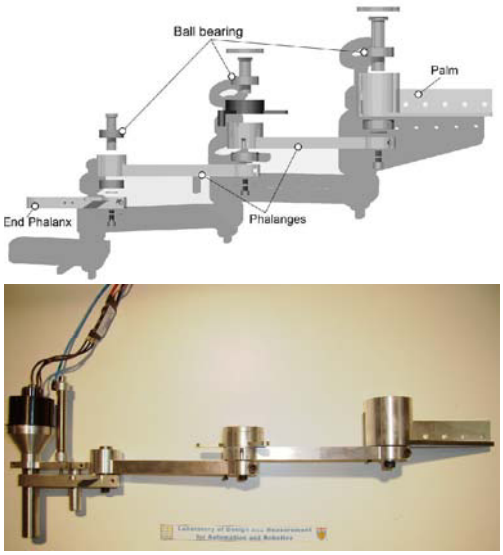


Figure 7 View of one lateral finger: digital and physical prototype

Figure 7 shows the lateral finger with the phalanges and grasping module mounted. Tests on the assembled finger stiffness and mechanical accuracy (backlash etc.) were performed on a specific testing bench including a Kuka KR5 robot.

The degree of freedom of the lateral fingers is 2 each one. The last link carrying the picking module can rotate around the last revolute joint and its orientation is commanded by a double rod. When the finger moves the phalanx is maintained parallel to the palm frame by these two rods. Part of the tuning of the finger dealt with the lengths of the two rods controlling finger orientation.

4.3.2 Central finger

Actuation is provided by an off the shelf linear actuator whose frame stands also as overall frame of the complete grasping device in order to save in overall mass.

4.3.3 Docking station

The docking station is the interface between the grasping device and the hanger. Four pneumatic actuators are used to realize all docking and locking operations including reset of the hanger after docking.

Figure 8 shows virtual and physical prototypes the docking unit.

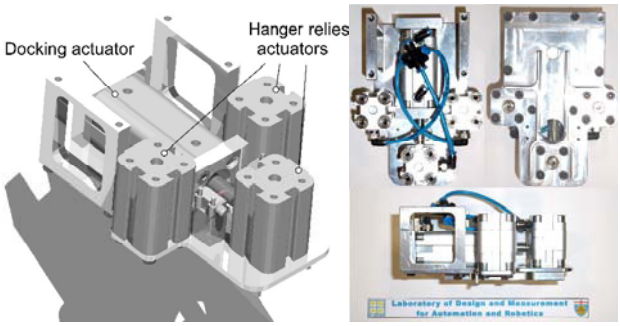


Figure 8 Docking unit virtual prototype (left) and physical prototype (right)

A flange is available on the top side of the gripper body for connection to the wrist of the robotic arm.

The ISO interface to the service robot is on the top, forming an angle with the plane of the fingers of the gripper, so that the service robot handles the grasping device closer to the middle of the angular span of its wrist.

4.3.5 Control implementation

One central control card is adopted to manage the eight motors in the fingers plus all on/off actuators and sensors in the grasping device. The DMC-40x0 is a full featured motion controller packaged with optional multi-axis drives in a compact, metal enclosure. The unit operates stand-alone or interfaces to a PC with Ethernet 10/100Base-T or RS232. The controller includes optically isolated I/O, high-power outputs capable of driving brakes or relays and analog inputs for interfacing to analog sensors. The DMC-40x0 controller and drive unit accepts power from a single 20–80 VDC source.

A commercial control card supporting the TCP/IP communication protocol was selected.. TCP/IP insures that if a

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packet is lost or destroyed while in transit, it will be resent. Ethernet communication transfers information in 'packets'. The packets are limited to 512 data bytes to avoid to loose the communication.

4.4 Handling system testing

After having tested stand alone the critical components the hanger and the grasping device were assembled, wired and embedded with cables and electronics. The first tests on the robotic system have been performed at the PMARlab in Genova in a test arrangement comprising, besides the grasping and hanging devices, a dummy cutting table with reduced functionalities but with real surface and jacket parts layout, a Kuka robot KR16 for driving the grasping device, a dummy docking station as reference position for the hanger, an emulator of the plant supervisor to communicate data to and from the cell control. **Figure 9** shows the test bench.



Figure 9 – Test bench at the PMARlab

After a very short set-up period and after having solved some problems about the communication and data formats translation, the first tests campaign started.
A further, more complete test campaign was performed within the Leapfrog pilot plant realized at the industrial premises of Robox in Italy to demonstrate the results of the project to textile/clothing industrialists. The pilot plant reconstructed a real challenging plant including the robotized resources and making them work on quality man clothes.

5 Conclusion

The results about the fabric picking efficiency and handling robustness, about the positioning repeatability on the cut parts reference points, about the on-line, real time reconfigurability of the grasping device, about the accuracy of the hanger catching and the ability to adjust it in air to the new configuration are really encouraging and let us hope in a near and quick technology transfer to the textile clothing manufacturing plants.

6 Acknowledgement

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